Intune Wi-Fi Profile Deployment & Management (Windows, iOS & Android)

Implementation Solution Guide

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Executive Summary

Corporate Wi-Fi should never be a manual step. Yet most organizations still hand users an SSID and a password, or leave proxy settings to guesswork — producing inconsistent connectivity, help-desk tickets, and security gaps from shared keys written on sticky notes. This solution replaces that with **centrally deployed Wi-Fi configuration profiles from Microsoft Intune**, delivered correctly to **Windows, iOS/iPadOS, and Android Enterprise** devices so each one connects automatically, with the right security type and the right proxy — no user action required.

The package is a **self-guided, click-by-click implementation** a competent administrator can follow end to end, plus five tested, read-only PowerShell scripts. It authors the Wi-Fi profile on all three platforms — handling the real differences in field labels, security-type options, and (critically) the **proxy field**, where a manual proxy is a **host + port** and only the automatic profile takes a **PAC URL**. Getting that one detail wrong is the single most common cause of broken Wi-Fi deployments; this guide gets it right per platform on the first try.

Delivery follows a disciplined **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model: every object is named to a consistent convention, assigned first to a named pilot group, proven against three explicit pass/fail gates (profile deployed, device auto-joins the SSID, proxy applied), then expanded to production with a documented rollback. For certificate-authenticated (WPA-Enterprise/802.1X) networks, the guide shows how to point the Wi-Fi profile at the certificate and RADIUS trust chain built by the companion Cisco ISE 802.1X product — deploying the profiles at scale without reproducing the PKI work.

The outcome: every managed device on every platform joins corporate Wi-Fi automatically and correctly — deployed once, validated on a pilot, and repeatable for every future rollout, at a fraction of the cost of a consulting engagement.

1. Solution Overview — Intune Wi-Fi Profile Deployment & Management (Windows, iOS & Android)

What this solution delivers

This solution deploys and manages **Wi-Fi configuration profiles** from **Microsoft Intune** to **Windows, iOS/iPadOS, and Android Enterprise** devices, so that every managed device connects to your corporate wireless network **automatically, correctly, and without the user typing an SSID, a password, or a proxy address**. You author the profile once per platform, assign it to a pilot group, prove that devices connect, then expand to the fleet.

In plain terms: instead of walking users through "join this SSID, here's the Wi-Fi password, now set the proxy," Intune pushes a policy that carries the network name, the security/authentication method, and the proxy configuration to the device silently. The Wi-Fi network appears pre-configured and the device joins it on its own.

What a Wi-Fi profile carries

A Wi-Fi configuration profile is a small policy object. Across all three platforms it always carries the same three decisions, and this solution deploys each one correctly per platform:

1. **The network (SSID).** The real broadcast name of the wireless network the device joins, plus a friendly **connection name** the user sees, and behavioral options — **connect automatically when in range** and whether the network is **hidden** (SSID not broadcast).

1. **The security / authentication type.** How the device proves it is allowed onto the network:

• **Open** — no authentication (guest/unsecured networks only).

• **Pre-shared key (WPA2/WPA3-Personal)** — one shared password/network key delivered by the profile. Simple; the same key goes to every targeted device.

• **Enterprise (WPA2/WPA3-Enterprise, 802.1X/EAP)** — the device authenticates with a certificate or credentials against a RADIUS server. This is the strongest option.

1. **The proxy.** Whether traffic on that network goes through a web proxy: **None**, **Manual** (a proxy server **address/host + port**), or **Automatic** (a **PAC-file URL**).

Where the enterprise / 802.1X-certificate path is handled

This product owns **Wi-Fi profile deployment and management at the operational level** — building, assigning, updating, and validating Wi-Fi profiles across the three platforms. It covers the **enterprise/802.1X security type at the deployment level**: selecting WPA2/WPA3-Enterprise, choosing an EAP type (EAP-TLS, PEAP, EAP-TTLS), and pointing the profile at an existing client-certificate and trusted-root profile.

It deliberately does **not** reproduce the deep certificate build. **Issuing the device client certificate (SCEP/PKCS), delivering the trusted root, and configuring the RADIUS authority (e.g. Cisco ISE) for EAP-TLS is a full solution of its own** — see the companion *Cisco ISE 802.1X Certificate Authentication for Wi-Fi & Wired via Intune* product. When your network is certificate-authenticated, deploy that solution to stand up the PKI + ISE + certificate profiles, then use **this** solution's method to author, assign, update, and validate the Wi-Fi profiles that reference those certificates at scale across Windows, iOS, and Android.

> ***Boundary in one line:*** *this product = *deploy and manage the Wi-Fi profiles across all platforms*. The ISE product = *build the certificate + RADIUS trust chain that a WPA-Enterprise profile authenticates against*.*

How Intune delivers the profile per platform

All three platforms use the same Intune template family (**Templates > Wi-Fi**), but the fields and options differ. This solution authors each platform correctly:

Platform	Profile family	Notable specifics
Windows 10/11	Templates > Wi-Fi (Basic or Enterprise)	Fields `Wi-Fi name (SSID)` + `Connection name`; `Connect automatically when in range`; `Company proxy settings`.

Platform	Profile family	Notable specifics
iOS/iPadOS	Templates > Wi-Fi (Basic or Enterprise)	Fields `Network name` + `SSID`; `Connect automatically`; `Hidden network`; proxy `Manual`/`Automatic`.
Android Enterprise	Templates > Wi-Fi (Basic or Enterprise)	Separate settings for **corporate-owned** vs **personally-owned work profile**; the work profile omits `Connect automatically` and manual proxy.

The proxy field — get it right (common failure)

The proxy configuration is where Wi-Fi profiles most often go wrong:

• **Manual proxy takes an address/host + a separate port number — it is NOT a URL.** You enter the

proxy server's host or IP address in one field and the port in another. Putting a scheme or a path (for example `http://proxy.contoso.com/proxy.pac`) into the manual **address** field breaks it — the whole string is treated as a hostname and fails to resolve.

• **Automatic proxy takes a PAC-file URL** (for example `http://proxy.contoso.com/proxy.pac`). The URL

belongs *only* in the automatic/PAC field, never in the manual address field.

This solution states, per platform, exactly which field is an address+port and which is a URL, so the proxy applies on the first try.

Naming model (named once, reused everywhere)

Every object this solution creates is named to a consistent convention — `WIFIMGT - <Purpose> - <Platform> - <Stage>` — and reused verbatim in every later step, in testing, and in the scripts:

Object (named)	What it is
`WIFIMGT - Pilot Devices`	Entra security group — the pilot assignment target.
`WIFIMGT - Production Devices`	Entra security group — the production assignment target.
`WIFIMGT - Corp WiFi - Windows - Pilot`	Windows Wi-Fi configuration profile.
`WIFIMGT - Corp WiFi - iOS - Pilot`	iOS/iPadOS Wi-Fi configuration profile.
`WIFIMGT - Corp WiFi - Android - Pilot`	Android Enterprise Wi-Fi configuration profile.

Benefits

• **Zero-touch connectivity** — devices join corporate Wi-Fi with no user setup.

- **Consistent, correct config** — the same SSID, security, and proxy settings on every device.
- **Cross-platform parity** — one repeatable method for Windows, iOS/iPadOS, and Android.
- **Safe change management** — a pilot-first, rollback-ready model for updating profiles later.

Limitations (be honest with buyers)

- **You can't target two Wi-Fi profiles with the same SSID to the same device** — a documented Intune

constraint; new profiles for the same network must use a distinct SSID or supersede the old one.

- **Changing the trusted root on an existing profile needs a backup internet path** on the device so the updated profile can be delivered (Android in particular).
- **Certificate (WPA-Enterprise/EAP-TLS) networks require the certificate + RADIUS chain first** — deploy the companion ISE product before the enterprise Wi-Fi profile will authenticate.
- **Profile changes take effect on the next device check-in**, not instantly — see the companion *"Wi-Fi changes not applying"* troubleshooting product for deep remediation.

The standard deployment model

> ***Assess*** → ***Configure*** → ***Pilot*** → ***Validate*** → ***Expand.*** *Confirm prerequisites and network facts → build the Wi-Fi profile on each platform → assign only to* `WIFIMGT - Pilot Devices` *→ prove devices connect with the right proxy → expand to* `WIFIMGT - Production Devices` *with a documented rollback.*

2. Requirements

Confirm every requirement below **before** authoring any Wi-Fi profile. Category-wide prerequisites (Entra tenant, Intune subscription, admin roles, enrollment, pilot group, network access) live in `global-requirements-intune-solutions.md`; this file adds the requirements **specific to Wi-Fi profile deployment across Windows, iOS/iPadOS, and Android Enterprise**. Each entry follows the house pattern: **Description · Why required · How to verify · How to implement if missing.**

R1. Known Wi-Fi network facts (SSID, security type, key/EAP)

Description: The exact broadcast **SSID**, the **security type** (Open, WPA2/WPA3-Personal with a pre-shared key, or WPA2/WPA3-Enterprise/802.1X), and — for Personal — the pre-shared key, or — for Enterprise — the EAP type and RADIUS server name. **Why required:** The profile reproduces these values exactly; a wrong SSID or key means no device ever connects. You can't target two profiles with the **same SSID** to the same device, so you must know the canonical SSID. **How to verify:** Confirm with the network/wireless team, or read the SSID from a working device (Windows: `netsh wlan show profiles`). **How to implement if missing:** Obtain the network design document or export a working profile from a reference device before proceeding.

R2. Proxy decision and values (per network)

Description: Whether the corporate Wi-Fi uses **no proxy**, a **manual proxy** (server **address/host + port**), or an **automatic proxy** (a **PAC-file URL**). **Why required:** Proxy is the most error-prone Wi-Fi field. The **manual** field is an address + a separate port (never a URL/path); the **automatic** field is a PAC URL. Choosing the wrong field, or pasting a URL into the manual address field, breaks connectivity. **How to verify:** Confirm the proxy model with the network team; on a working Windows device check **Settings > Network & Internet > Proxy**. **How to implement if missing:** Get the proxy host + port, or the PAC URL, from the network team. If no proxy is used, set proxy to **None** deliberately.

R3. Wi-Fi profile authoring role (Policy and Profile Manager)

Description: An Intune account with the **Policy and Profile Manager** built-in role (or Intune Administrator). **Why required:** Creating and assigning Wi-Fi configuration profiles requires this role at minimum. **How to verify:** Intune admin center → **Tenant administration > Roles**; confirm the operator holds **Policy and Profile Manager** or **Intune Administrator**. **How to implement if missing:** Assign the **Policy and Profile Manager** role to the operator.

R4. Enrolled devices on each target platform

Description: The Windows, iOS/iPadOS, and/or Android Enterprise devices you intend to configure are **enrolled in Intune**, in the management modes you plan to target. **Why required:** Wi-Fi profiles only reach enrolled, managed devices; Android options differ between **corporate-owned** (fully managed / dedicated / COPE) and **personally-owned work profile** (BYOD). **How to verify:** Intune → **Devices > All devices**; confirm target devices appear and are compliant. Note each Android device's ownership/management mode. **How to implement if missing:** Complete enrollment for each platform (Windows Autopilot / automatic enrollment; Apple ADE or user enrollment; Android Enterprise enrollment) before assigning Wi-Fi.

R5. Apple MDM push certificate (iOS/iPadOS)

Description: A valid **Apple MDM Push certificate (APNs)** connected to the tenant. **Why required:** No Apple device configuration profile — including Wi-Fi — deploys without a valid APNs certificate; an expired APNs silently stops all iOS/iPadOS policy. **How to verify:** Intune → **Devices > Enrollment > Apple > Apple MDM Push certificate**; confirm it is present and **not expired**. **How to implement if missing:** Renew/create the APNs certificate (Apple Business Manager account is needed for ADE, but a push certificate alone suffices for user-enrolled devices).

R6. Managed Google Play connection (Android Enterprise)

Description: The tenant is connected to **managed Google Play** (Android Enterprise binding). **Why required:** Android Enterprise Wi-Fi profiles (fully managed, dedicated, COPE, or work profile) require the Android Enterprise connection to exist. **How to verify:** Intune → **Devices > Enrollment > Android > Managed Google Play**; confirm status is connected. **How to implement if missing:** Complete the managed Google Play binding (see the companion Android Enterprise connection solution) before authoring Android Wi-Fi profiles.

R7. Certificate + RADIUS trust chain (ONLY if using WPA-Enterprise / 802.1X)

Description: If the network security type is **Enterprise (WPA2/WPA3-Enterprise, EAP-TLS/PEAP)**, the device must already have a **client certificate profile** (SCEP or PKCS), a **trusted-root profile** for RADIUS-server validation, and a working **RADIUS authority** (e.g. Cisco ISE). **Why required:** An enterprise Wi-Fi profile references — it does not create — these certificate objects. Without them the profile deploys but authentication fails. **How to verify:** Intune → **Devices > Configuration**; confirm a trusted-root and a SCEP/PKCS cert profile exist and are assigned to the same devices; confirm RADIUS accepts EAP for the SSID. **How to implement if missing:** Deploy the companion **Cisco ISE 802.1X Certificate Authentication for Wi-Fi & Wired via Intune** solution to build the PKI, certificate profiles, and ISE policy first. For **Personal (pre-shared key)** or **Open** networks, this requirement does **not** apply — skip it.

R8. Pilot and production assignment groups

Description: Two Entra security groups: `WIFIMGT - Pilot Devices` and `WIFIMGT - Production Devices` (device groups, since Wi-Fi config is typically device-targeted). **Why required:** Every profile is piloted before fleet rollout, then expanded — separate groups make that safe and reversible. **How to verify:** Entra → **Groups**; confirm both groups exist. **How to implement if missing:** Create them (Entra → **Groups > New group > Security**), then add a few known test devices to `WIFIMGT - Pilot Devices`.

R9. Government cloud awareness (GCC High / DoD tenants only)

Description: If you operate in a **US Government** tenant, use the sovereign-cloud portals and Graph endpoints throughout. **Why required:** Government tenants use different admin URLs and Graph endpoints; the steps otherwise match Commercial for Wi-Fi profiles (see the note in the Implementation guide). **How to verify:** You sign in at `https://intune.microsoft.us` (not `.com`). **How to implement if missing:** Substitute `intune.microsoft.us` / `graph.microsoft.us` and connect scripts with `Connect-MgGraph -Environment USGov` (or `USGovDoD`). No Wi-Fi-profile field differs materially; **verify for GCC High** where your tenant's feature rollout lags Commercial.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation

Follow these steps in order. You will create two assignment groups, then author one Wi-Fi profile per platform (Windows, then iOS/iPadOS, then Android Enterprise), assign each to the pilot group, and later expand to production. Every object is named exactly as shown and the name is reused verbatim later.

All navigation is for the **Microsoft Intune admin center** (`https://intune.microsoft.com`). **US Government tenants:** use `https://intune.microsoft.us` instead — every menu path below is identical; only the host differs (see Step 0 note).

> ***Step 0 — Government cloud note.*** *In a GCC High / DoD tenant, sign in at* `intune.microsoft.us`*; the scripts connect with* `Connect-MgGraph -Environment USGov` *(or* `USGovDoD`*). No Wi-Fi-profile field below changes for Government. Where your tenant's feature rollout lags Commercial,* ***verify for GCC High****.*

Phase 1 — Create the assignment groups

Step 1. Create the pilot group

1. Go to `https://entra.microsoft.com` and sign in as an administrator.
2. Select **Groups** > **All groups** > **New group**.
3. Set **Group type** = **Security**.
4. **Group name**: enter `WIFIMGT - Pilot Devices`.
5. **Membership type**: **Assigned**.
6. Select **No members selected** > add 3–5 known test devices (one Windows, one iOS/iPadOS, one Android where possible) > **Select**.
1. Select **Create**.

Step 2. Create the production group

1. Still in **Groups** > **All groups** > **New group**.
2. **Group type** = **Security**; **Group name** = `WIFIMGT - Production Devices`; **Membership type** = **Assigned** (or Dynamic, per your fleet convention).
1. Select **Create**. Leave it empty for now — you populate it at expansion (Phase 5).

Phase 2 — Author the Windows Wi-Fi profile

Step 3. Start the Windows profile

1. In the Intune admin center, select **Devices** > **Manage devices** > **Configuration** > **Create** > **New policy**.
1. **Platform**: select **Windows 10 and later**.
2. **Profile type**: select **Templates**, then choose **Wi-Fi** > **Create**.

Step 4. Name the Windows profile

1. On the **Basics** tab, **Name**: enter `WIFIMGT - Corp WiFi - Windows - Pilot`.
2. **Description**: enter `Corporate Wi-Fi for Windows devices — pilot ring.`
3. Select **Next**.

Step 5. Configure the Windows Wi-Fi settings

On the **Configuration settings** tab:

1. **Wi-Fi type**:

• For a pre-shared-key network, select **Basic**.

• For a certificate/802.1X network, select **Enterprise** (see Step 6).

1. **Wi-Fi name (SSID)**: enter the real broadcast SSID exactly, e.g. `Contoso-Corp`.
2. **Connection name**: enter the friendly name users see, e.g. `Contoso Corp Wi-Fi`.
3. **Connect automatically when in range**: **Yes**.

• **Connect to more preferred network if available**: **No** (keep devices on the corporate SSID).

• **Connect to this network, even when it is not broadcasting its SSID**: set **Yes** only if the SSID is hidden; otherwise **No**.

1. **Wireless Security Type** (Basic):

• **Open (no authentication)** — guest/unsecured networks only, or

• **WPA/WPA2-Personal** — then enter the **Pre-shared key** (an 8–63 character ASCII string or 64 hex characters). The PSK is identical on every targeted device — keep it secure.

1. **Company proxy settings** — choose one and enter values in the correct field:

• **None** — no proxy.

• **Manually configure** — enter the **Proxy server IP address** (host/IP only, no scheme, no path) and its **Port number** (e.g. address `10.0.0.22`, port `8080`).

• **Automatically configure** — enter the **PAC-file URL**, e.g. `http://proxy.contoso.com/proxy.pac`.

(This is the only proxy field that takes a URL.)

1. Select **Next**.

Step 6. (Enterprise/802.1X networks only) Windows Enterprise settings

If you chose **Enterprise** in Step 5, also configure:

1. **Authentication mode**: **User or machine** (typical), or **Machine** for unattended devices.
2. **EAP type**: select **EAP-TLS** for certificate authentication (recommended). PEAP and EAP-TTLS are also available for credential-based auth.

1. For **EAP-TLS**:

• **Certificate server names**: enter the RADIUS server certificate common name(s).

• **Root certificates for server validation**: select the existing **trusted-root** profile.

• **Authentication method**: select **SCEP certificate** (or **PKCS certificate**) and choose the existing client-certificate profile.

> *These certificate objects are **not** created here — they come from the companion Cisco ISE 802.1X solution (see Requirement R7). This profile only *references* them.*

1. **Force Wi-Fi profile to be compliant with FIPS**: set **Yes** only if your agency requires FIPS 140-2 validation; otherwise **No**.

Step 7. Scope tags, assign, and create (Windows)

1. On **Scope tags** (optional): add a scope tag if your org uses them > **Next**.
2. On **Assignments**: under **Included groups**, select **Add groups**, choose `WIFIMGT - Pilot Devices`, then **Next**.
1. On **Review + create**, confirm the SSID, security type, and proxy are correct > **Create**.

Phase 3 — Author the iOS/iPadOS Wi-Fi profile

Step 8. Start the iOS/iPadOS profile

1. Select **Devices** > **Manage devices** > **Configuration** > **Create** > **New policy**.
2. **Platform**: select **iOS/iPadOS**.
3. **Profile type**: select **Templates**, then choose **Wi-Fi** > **Create**.

Step 9. Name the iOS/iPadOS profile

1. **Basics** > **Name**: enter `WIFIMGT - Corp WiFi - iOS - Pilot`.
2. **Description**: enter `Corporate Wi-Fi for iOS/iPadOS devices — pilot ring.`
3. Select **Next**.

Step 10. Configure the iOS/iPadOS Wi-Fi settings

On **Configuration settings**:

1. **Wi-Fi type**: **Basic** (pre-shared key) or **Enterprise** (802.1X — see Step 11).
2. **Network name**: enter the friendly connection name, e.g. `Contoso Corp Wi-Fi`.
3. **SSID**: enter the real broadcast SSID exactly, e.g. `Contoso-Corp`.
4. **Connect automatically**: **Enable**.
5. **Hidden network**: **Enable** only if the SSID is not broadcast; otherwise **Disable**.
6. **Security type** (Basic):

• **Open (no authentication)**, or

• **WPA2/WPA3 - Personal** (recommended) / **WPA/WPA2 - Personal** — then enter the **Pre-shared key**.

1. **Proxy settings**:

• **None** — no proxy.

• **Manual** — enter the **Proxy server address** (IP address, no scheme/path) and its **Port number**.

• **Automatic** — enter the **Proxy server URL** that points to the PAC file, e.g. `http://proxy.contoso.com/proxy.pac`.

1. **Disable MAC address randomization**: leave **Not configured** unless NAC requires the hardware

MAC (setting applies to iOS/iPadOS 14.0+).

1. Select **Next**.

Step 11. (Enterprise/802.1X networks only) iOS/iPadOS Enterprise settings

If you chose **Enterprise** in Step 10:

1. **Security type**: **WPA/WPA2 - Enterprise** (or **WPA - Enterprise**).
2. **EAP type**: select **EAP-TLS** (recommended). EAP-TTLS, PEAP, EAP-FAST, EAP-SIM, and LEAP are

also offered.

1. For **EAP-TLS**:

• **Certificate server names**: **Add** the RADIUS server certificate common name(s), e.g.

`radius.contoso.com` or a wildcard `*.contoso.com`.

• **Root certificate for server validation**: select the existing trusted-root profile.

• **Authentication method** > **Certificates**: select the existing SCEP or PKCS client-certificate

profile.

> *As with Windows, these certificate objects come from the companion Cisco ISE 802.1X solution; this profile only references them.*

Step 12. Scope tags, assign, and create (iOS/iPadOS)

1. **Scope tags** (optional) > **Next**.
2. **Assignments** > **Add groups** > `WIFIMGT - Pilot Devices` > **Next**.
3. **Review + create** > confirm SSID/security/proxy > **Create**.

Phase 4 — Author the Android Enterprise Wi-Fi profile

> *Android options differ by ownership. Steps 13–16 cover **corporate-owned** (fully managed, dedicated, COPE). For **personally-owned work profile (BYOD)**, see Step 17 for the differences.*

Step 13. Start the Android Enterprise profile

1. Select **Devices** > **Manage devices** > **Configuration** > **Create** > **New policy**.
2. **Platform**: select **Android Enterprise**.
3. Under **Templates**, select the profile type matching your devices:

• **Fully Managed, Dedicated, and Corporate-Owned Work Profile** > **Wi-Fi**, or

• **Personally-Owned Work Profile** > **Wi-Fi**.

1. Select **Create**.

Step 14. Name the Android profile

1. **Basics** > **Name**: enter `WIFIMGT - Corp WiFi - Android - Pilot`.

2. **Description**: enter `Corporate Wi-Fi for Android Enterprise devices — pilot ring.`

3. Select **Next**.

Step 15. Configure the Android (corporate-owned) Wi-Fi settings

On **Configuration settings**:

1. **Wi-Fi type**: **Basic** (pre-shared key) or **Enterprise** (802.1X — see Step 16).

2. **Network name**: enter the friendly name, e.g. `Contoso Corp Wi-Fi`.

3. **SSID**: enter the real broadcast SSID exactly, e.g. `Contoso-Corp`.

*Reminder: you can't target two profiles with the **same SSID** to one device.*

1. **Connect automatically**: **Enable**.

2. **Hidden network**: **Enable** only if the SSID is not broadcast; otherwise **Disable**.

3. **Security type** (Basic):

• **Open (no authentication)**, or

• **WPA-Pre-shared key** (recommended; WEP is deprecated on Android 12+) — then enter the **Pre-shared key**.

1. **Proxy settings**:

• **None** — no proxy.

• **Manual** — enter the **Proxy server address** (IP address), the **Port number**, and an optional **Exclusion list** (one host/IP per line; * wildcard allowed).

• **Automatic** — enter the **Proxy server URL** pointing to the PAC file, e.g. `http://proxy.contoso.com/proxy.pac`.

1. **MAC address randomization** (Android 13+): leave **Use device default** unless NAC needs the hardware MAC (then **Use device MAC**).

1. Select **Next**.

Step 16. (Enterprise/802.1X networks only) Android Enterprise settings

If you chose **Enterprise** in Step 15:

1. **EAP type**: select **EAP-TLS** (recommended); EAP-TTLS and PEAP are also available.

2. For **EAP-TLS**:

• **Radius server name**: enter the DNS name in the RADIUS server certificate (e.g. `contoso.com`). On Android 11+ this is often required or the device won't connect; on Android 14+ keep the total under 256 characters (use the suffix only).

• **Root certificate for server validation**: select the existing trusted-root profile.

• **Authentication method** > **Certificates**: select the existing SCEP or PKCS client-certificate profile.

- **Identity privacy (outer identity)**: optionally enter `anonymous`.

*For **personally-owned work profile** EAP, the client certificate's **Subject Alternative Name (SAN) must include the UPN**, or the Wi-Fi profile deployment fails.*

Step 17. Personally-owned work profile (BYOD) differences

If you selected **Personally-Owned Work Profile > Wi-Fi** in Step 13, note these differences from the corporate-owned steps:

- There is **no Connect automatically** setting (the work profile manages that).
- **Proxy settings** offer only **None** or **Automatic** (a **Proxy server URL**) — **Manual proxy is not available**. If your BYOD network needs a proxy, you must use a PAC URL.
- Otherwise SSID, Hidden network, security type / EAP, and certificate references match Step 15/16.

Step 18. Scope tags, assign, and create (Android)

1. **Scope tags** (optional) > **Next**.
2. **Assignments** > **Add groups** > `WIFIMGT - Pilot Devices` > **Next**.
3. **Review + create** > confirm SSID/security/proxy > **Create**.

Phase 5 — Validate the pilot, then expand

Step 19. Validate on pilot devices

Run the tests in **Testing & Validation** (`05-testing.md`) on at least one device per platform: the profile shows **Succeeded** in Intune, the device joins the SSID automatically, and (if a proxy is configured) traffic honors the proxy. Do not expand until all pilot gates pass.

Step 20. Expand to production

1. Add the fleet devices to `WIFIMGT - Production Devices` (or point its dynamic rule at them).
2. For each of the three profiles (`WIFIMGT - Corp WiFi - Windows - Pilot`, `... - iOS - Pilot`, `... - Android - Pilot`), open the profile > **Properties** > **Assignments**

***Edit**, and **add** `WIFIMGT - Production Devices` to **Included groups**. Keep the pilot group* assigned so pilot devices remain covered. Select **Review + save**.

Optional: rename the profiles from `- Pilot` to `- Prod` once the pilot ring is retired, and update the scripts' `-ProfileNames` accordingly.

Step 21. Rollback (if needed)

To reverse an expansion, open the affected profile > **Assignments** > **Edit**, **remove** `WIFIMGT - Production Devices`, and **Review + save**. Devices drop the pushed Wi-Fi profile on their next check-in. For a full stop, set the profile's assignment to **Excluded** for the affected group.

5. Testing & Validation

Validate on the pilot ring before expanding. There are three gates per device: **(A) the profile deployed**, **(B) the device joined the SSID automatically**, and **(C) the proxy applied correctly**. Run gate A from the Intune console + scripts; run gates B and C on the device itself. Do **not** expand to `WIFIMGT - Production Devices` until every gate PASSes on at least one device per platform.

Gate A — The Wi-Fi profile deployed (console + Graph)

Console check. Intune admin center > **Devices** > **Manage devices** > **Configuration** > open `WIFIMGT - Corp WiFi - Windows - Pilot` (repeat for iOS and Android) > **Device status**.

PASS	FAIL
Each pilot device shows **Succeeded**.	Any pilot device shows **Error** / **Conflict** / **Not applicable**.

- **Conflict** almost always means a second Wi-Fi profile targets the **same SSID** on that device — remove the duplicate (see Troubleshooting).
- **Not applicable** usually means the device is the wrong platform/ownership for the profile.

Script check. Run the assignment validator (read-only):

```
powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Validate-Assignment.ps1 -ProfileNames `
  'WIFIMGT - Corp WiFi - Windows - Pilot','WIFIMGT - Corp WiFi - iOS - Pilot','WIFIMGT - Corp WiFi - Andro
```

PASS: each profile lists `WIFIMGT - Pilot Devices` as an assigned group and the expected device count. **FAIL:** a profile is UNASSIGNED or missing the pilot group.

Gate B — The device joins the SSID automatically

Windows. On a pilot Windows device (do not manually type the password):

1. Open **Settings > Network & Internet > Wi-Fi** — the corporate SSID appears and connects.
2. Confirm the pushed profile exists: `netsh wlan show profiles` lists the SSID.
3. Confirm it is connected: `netsh wlan show interfaces` shows **State : connected** to the SSID.

PASS	FAIL
Device auto-connects to the SSID; profile present via `netsh`.	SSID absent from `show profiles`, or device won't associate.

iOS/iPadOS. On a pilot device: **Settings > General > VPN & Device Management** shows the management profile; **Settings > Wi-Fi** shows the corporate SSID connected without the user entering a key. **PASS:** connected automatically. **FAIL:** SSID not present or prompts for a password.

Android Enterprise. On a pilot device: **Settings > Network & internet > Internet** shows the corporate SSID connected. **PASS:** connected automatically. **FAIL:** SSID absent or shows "authentication error" (for Enterprise, check the certificate/RADIUS chain — Requirement R7).

Gate C — The proxy applied correctly

Only if a proxy is configured in the profile.

Windows.

- **Manual proxy:** `netsh winhttp show proxy` (or **Settings > Network & Internet > Proxy**) shows the

host + port you entered — **not** a URL. Confirm the host resolves and the port is reachable.

- **Automatic (PAC):** the PAC URL is shown under **Automatic proxy setup**; browse to an internal site

and confirm it routes via the proxy.

PASS	FAIL
Proxy host+port (manual) or PAC URL (automatic) is present and traffic honors it.	Manual field shows a URL/path (misconfigured), or DNS fails to resolve the "host".

> ***Key check:*** *if the manual proxy "address" contains a scheme (`http://`) or a path (`/proxy.pac`), it is wrong — the whole string is treated as a hostname and fails. Move a PAC URL to the* ***Automatic*** *field.*

iOS/iPadOS & Android. Load an internal-only resource that requires the proxy; it succeeds via the proxy. For manual proxy, confirm the address is a bare host/IP and the port is separate; for automatic, confirm the PAC URL loads.

Whole-fleet coverage snapshot

Before and after expansion, run the connectivity/coverage report (read-only):

```
powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Report-Connectivity.ps1
```

PASS: reported deployment-state counts for the three profiles trend to **Succeeded** across the assigned devices with few/no Errors. **FAIL:** a rising Error/Conflict count — investigate before widening assignment.

Exit criteria (all must be true to expand)

- Gate A: all three profiles **Succeeded** on their pilot devices (console + script).
- Gate B: one device per platform auto-joins the SSID.
- Gate C: proxy (if any) applied correctly — manual = host+port, automatic = PAC URL.
- `WIFIMGT-Report-Connectivity.ps1` shows no unexplained Errors/Conflicts.

6. Troubleshooting

Quick fixes for the common Wi-Fi profile deployment failures. For **deep "the profile deployed but the change isn't reflected on the device" diagnosis** (sync timing, per-setting conflicts, remediation of stuck Wi-Fi changes), use the companion **Intune Wi-Fi Profile Remediation — "changes not applying"** troubleshooting product rather than reproducing that full diagnostic here.

Symptom	Likely cause	Fix
Profile shows **Conflict** on a device	Two Wi-Fi profiles target the **same SSID** on that device	Intune won't target two same-SSID profiles to one device. Remove/supersede the duplicate so only one active profile carries that SSID.
Profile shows **Not applicable**	Wrong platform or Android ownership mode for the profile	Confirm the device platform; for Android, confirm corporate-owned vs personally-owned work profile matches the profile type you created.
Device won't associate to the SSID	Wrong SSID or wrong pre-shared key	Re-check the **SSID** (exact broadcast name) and the **Pre-shared key** against the network facts; a single wrong character fails silently.
Proxy not working / no internet on the network	A URL/path was entered in the **manual** proxy address field	Manual proxy = **host/IP + port only** (no `http://`, no `/path`). Move any PAC URL to the **Automatic** field.
Automatic proxy ignored	PAC URL unreachable or malformed	Confirm the **PAC-file URL** loads from the device's network and returns a valid PAC script.
Enterprise (802.1X) auth fails though the profile deployed	Missing/mismatched certificate or trusted-root, or RADIUS not accepting EAP	Verify the client-cert and trusted-root profiles are assigned to the same devices and the RADIUS (e.g. Cisco ISE) policy accepts EAP for the SSID — see the companion **Cisco ISE 802.1X** solution.
Android work-profile EAP profile fails to deploy	Client certificate **SAN lacks the UPN**	Reissue the certificate so its Subject Alternative Name includes the user principal name (UPN).
Android device shows no manual proxy option	It's a **personally-owned work profile**	Manual proxy is unavailable on BYOD work profiles — use an **Automatic (PAC URL)** proxy instead.
Changed **trusted root**, devices dropped off Wi-Fi	No backup internet path to deliver the updated profile	Ensure devices have a second internet connection (e.g. cellular/another SSID) so the updated profile lands, then reconnect.

Symptom	Likely cause	Fix
Change made in the console but device still on old settings	Device hasn't checked in yet	Wait for the next sync (or force a sync); if it persists, use the companion **"changes not applying"** troubleshooting product.
`WIFIMGT-*` script errors on connect	Wrong cloud environment or missing scopes	Re-run with `-Environment USGov/USGovDoD` for Government tenants; consent to the read scopes when prompted.

First diagnostic pass

1. Run `WIFIMGT-Validate-Assignment.ps1` — confirm the named profile is assigned to the pilot/prod

group.

1. Run `WIFIMGT-Report-Connectivity.ps1` — see the Succeeded/Error/Conflict split across the fleet.

2. On one failing device, walk **Gate B** and **Gate C** from `05-testing.md` (device-side `netsh`

checks on Windows; Settings on iOS/Android).

1. If the profile is assigned and shows **Succeeded** but the device behavior is wrong, hand off to the

companion **"changes not applying"** troubleshooting product.

7. Best Practices

Naming and hygiene

• Keep the `WIFIMGT - <Purpose> - <Platform> - <Stage>` convention on every object and reuse the exact

name in profiles, assignments, testing, and scripts. It makes audit, handoff, and cleanup trivial.

• One SSID → one active profile per device. Because Intune **won't target two profiles with the same

SSID to the same device**, never leave an old and a new profile for the same network both assigned — supersede and remove the old one.

Security type

• Prefer **WPA2/WPA3-Personal** over open networks; prefer **WPA2/WPA3-Enterprise (EAP-TLS)** over

pre-shared keys wherever a certificate + RADIUS chain exists — a leaked PSK is the same on every device you deployed it to.

• Treat the **pre-shared key as a secret**: rotate it on a schedule and whenever a device is lost.

Changing the PSK means updating and redeploying the profile.

• For certificate (Enterprise) networks, deploy the **Wi-Fi profile, client-certificate profile, and trusted-root profile to the same groups** so every device can validate the chain — a Microsoft-recommended practice. The certificate build itself belongs to the companion Cisco ISE 802.1X product.

Proxy

• Decide proxy per network **before** authoring, and put the value in the right field: **Manual = host/IP + port **(no scheme, no path);** Automatic = PAC URL**. This single discipline prevents the most common Wi-Fi failure.

• On Android **personally-owned work profiles**, manual proxy is unavailable — standardize on a **PAC URL** for BYOD networks that need a proxy.

Change management

• Always **pilot first** (`WIFIMGT - Pilot Devices`), validate the three gates, then expand to `WIFIMGT - Production Devices`. Keep the pilot assignment in place after expansion.

• Before changing a **trusted root** on an existing enterprise profile, make sure devices have a **backup internet path** so the updated profile can be delivered (critical on Android).

• Profile edits apply on the **next device check-in**, not instantly. Don't conclude a change "didn't work" until devices have synced — for stuck changes, use the companion *Wi-Fi changes not applying* troubleshooting product.

• Document every change (what SSID/security/proxy value changed, when, and the rollback) so an expansion can be reversed cleanly.

Platform parity

• Author all three platforms from the **same network facts** so a user on any device gets the same SSID, security, and proxy behavior.

• Re-verify field labels against current Microsoft Learn documentation before large rollouts — Intune Wi-Fi templates evolve, and the Settings Catalog exposes additional Wi-Fi settings not in the templates.

Validation as a habit

• Run `WIFIMGT-Report-Connectivity.ps1` on a schedule to catch profiles drifting into Error/Conflict as the fleet changes.

• Keep `WIFIMGT-Remediation.ps1` in its default **-WhatIf** preview posture; only pass `-Apply` after reviewing exactly which assignments it will add.

8. Deliverables Summary

This solution is delivered as self-guided files you keep and reuse across unlimited devices.

Documentation (Word + PDF)

• **Solution Overview** — what a Wi-Fi profile carries (SSID, security type, proxy), the cross-platform delivery model, the boundary vs the Cisco ISE 802.1X certificate product, and the naming model.

• **Requirements Guide** — Wi-Fi-specific prerequisites (network facts, proxy decision, roles, per-platform enrollment, APNs, managed Google Play, the certificate/RADIUS chain for Enterprise, pilot/production groups, Government cloud) each with Verify / Implement-if-missing.

• **Implementation Guide** — click-by-click deployment of the Wi-Fi profile on **Windows**, **iOS/iPadOS**, and **Android Enterprise** (corporate-owned and BYOD work profile), in order, with every object named and assigned pilot → production, plus rollback.

• **Testing & Validation Guide** — three pass/fail gates (profile deployed, device auto-joins the SSID, proxy applied correctly) per platform, with device-side commands.

• **Troubleshooting Guide** — the common Wi-Fi deployment failures and fixes (deep "changes not applying" diagnosis is deferred to the companion troubleshooting product).

• **Best Practices** — naming, security-type choice, proxy discipline, and safe change management.

PowerShell scripts (ZIP) — five, read-only by default, cloud-aware

All connect with `Connect-MgGraph -Environment Global|USGov|USGovDoD` for Commercial or US Government tenants:

• `WIFIMGT-Preflight.ps1` — checks tenant readiness for Wi-Fi profile deployment (SDK, Graph scopes, APNs, Android Enterprise binding, pilot group).

• `WIFIMGT-Get-WifiProfiles.ps1` — exports existing Wi-Fi profiles across platforms with their assignments.

• `WIFIMGT-Validate-Assignment.ps1` — reports which groups/devices received the named Wi-Fi profiles.

• `WIFIMGT-Report-Connectivity.ps1` — Wi-Fi profile deployment-state (Succeeded/Error/Conflict) across the fleet.

• `WIFIMGT-Remediation.ps1` — guarded (-WhatIf by default) fix for missing pilot/production group assignments on the named profiles.

Package

Everything above is delivered together as `WIFIMGT-Complete-Solution.zip` (the Word document, the PDF, and all PowerShell scripts).

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WIFIMGT-Get-WifiProfiles.ps1

Purpose: Lists every Wi-Fi device-configuration profile in the tenant across Windows, iOS/iPadOS, and Android Enterprise (template-based, identified by @odata.type) plus any Settings Catalog policies whose name suggests Wi-Fi, and shows each profile's assigned groups. Use it to inventory what exists before authoring new profiles (and to spot same-SSID duplicates). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Get-WifiProfiles.ps1 Sovereign cloud: .\WIFIMGT-Get-WifiProfiles.ps1 -Environment USGov (or USGovDoD) Export to CSV: .\WIFIMGT-Get-WifiProfiles.ps1 -OutputCsv .\wifi-profiles.csv Required Graph scopes: DeviceManagementConfiguration.Read.All EXPECTED RESULTS A table of Wi-Fi profiles (name, platform/type, assigned groups). Read-only — nothing changes. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string] $OutputCsv)

How to run: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Get-WifiProfiles.ps1 Sovereign cloud: .\WIFIMGT-Get-WifiProfiles.ps1 -Environment USGov (or USGovDoD) Export to CSV: .\WIFIMGT-Get-WifiProfiles.ps1 -OutputCsv .\wifi-profiles.csv Required Graph scopes: DeviceManagementConfiguration.Read.All EXPECTED RESULTS A table of Wi-Fi profiles (name, platform/type, assigned groups). Read-only — nothing changes. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string] $OutputCsv)

WIFIMGT-Preflight.ps1

Purpose: Confirms the tenant is ready to deploy Wi-Fi configuration profiles: the Microsoft.Graph SDK is available, Graph connects with the required read scopes, an Apple MDM Push (APNs) certificate is present (for iOS/iPadOS), the Android Enterprise (managed Google Play) binding exists, and the pilot assignment group is present. Nothing is changed. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Preflight.ps1 Sovereign cloud: .\WIFIMGT-Preflight.ps1 -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementServiceConfig.Read.All, Group.Read.All EXPECTED RESULTS A PASS/GAP checklist and an overall verdict. Read-only against Microsoft Graph. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string] $PilotGroup = 'WIFIMGT - Pilot Devices')

How to run: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Preflight.ps1 Sovereign cloud: .\WIFIMGT-Preflight.ps1 -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementServiceConfig.Read.All, Group.Read.All EXPECTED RESULTS A PASS/GAP checklist and an overall verdict. Read-only against Microsoft Graph. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string] $PilotGroup = 'WIFIMGT - Pilot Devices')

WIFIMGT-Remediation.ps1

Purpose: Closes assignment gaps found by WIFIMGT-Validate-Assignment.ps1 by ensuring each named Wi-Fi profile is assigned to the intended target group. It ONLY adds a group assignment where one is missing; it never deletes an assignment, never edits Wi-Fi settings (SSID/security/proxy), and never touches any other profile. Runs in preview (-WhatIf) by default — pass -Apply to commit. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Remediation.ps1 -TargetGroup 'WIFIMGT - Pilot Devices' Apply: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Remediation.ps1 -TargetGroup 'WIFIMGT - Pilot Devices' -Apply Sovereign cloud: add -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.ReadWrite.All, Group.Read.All EXPECTED RESULTS A list of profiles that are missing the target-group assignment and, when -Apply is set, confirmation that the assignment was added. Existing assignments are preserved. .NOTES White-label — Intune Solutions. Guarded write against Microsoft Graph (default preview / -WhatIf). > [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [Parameter(Mandatory)][string] $TargetGroup, [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'), [switch] $Apply)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Remediation.ps1 -TargetGroup 'WIFIMGT - Pilot Devices' Apply: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Remediation.ps1 -TargetGroup 'WIFIMGT - Pilot Devices' -Apply Sovereign cloud: add -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.ReadWrite.All, Group.Read.All EXPECTED RESULTS A list of profiles that are missing the target-group assignment and, when -Apply is set, confirmation that the assignment was added. Existing assignments are preserved. .NOTES White-label — Intune Solutions. Guarded write against Microsoft Graph (default preview / -WhatIf). > [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [Parameter(Mandatory)][string] $TargetGroup, [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'), [switch] $Apply)

WIFIMGT-Report-Connectivity.ps1

Purpose: For the named Wi-Fi profiles, reports the per-device deployment state (Succeeded / Error / Conflict / Pending / Not applicable) using Intune's deviceConfiguration deviceStatuses, so you can see whether devices actually received the profile. A rising Error/Conflict count is the signal to investigate before widening assignment. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Report-Connectivity.ps1 Sovereign cloud: .\WIFIMGT-Report-Connectivity.ps1 -Environment USGov (or USGovDoD) Export to CSV: .\WIFIMGT-Report-Connectivity.ps1 -OutputCsv .\wifi-state.csv Required Graph scopes: DeviceManagementConfiguration.Read.All EXPECTED RESULTS A per-profile state summary (counts by status) and, with -OutputCsv, a per-device breakdown. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'), [string] $OutputCsv)

How to run: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Report-Connectivity.ps1 Sovereign cloud: .\WIFIMGT-Report-Connectivity.ps1 -Environment USGov (or USGovDoD) Export to CSV: .\WIFIMGT-Report-Connectivity.ps1 -OutputCsv .\wifi-state.csv Required Graph scopes: DeviceManagementConfiguration.Read.All EXPECTED RESULTS A per-profile state summary (counts by

status) and, with -OutputCsv, a per-device breakdown. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'), [string] $OutputCsv)

WIFIMGT-Validate-Assignment.ps1

Purpose: For each named Wi-Fi profile, confirms it exists in Intune and lists the groups it is assigned to (and the member count of each group), so you can prove the pilot/production targeting is correct before and after expansion. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Validate-Assignment.ps1 Named profiles: .\WIFIMGT-Validate-Assignment.ps1 -ProfileNames 'WIFIMGT - Corp WiFi - Windows - Pilot','WIFIMGT - Corp WiFi - iOS - Pilot' Sovereign cloud: .\WIFIMGT-Validate-Assignment.ps1 -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.Read.All, Group.Read.All EXPECTED RESULTS Per profile: FOUND/MISSING, the assigned group name(s), and each group's member count. A profile with no assignment is flagged UNASSIGNED. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'))

How to run: powershell -ExecutionPolicy Bypass -File .\WIFIMGT-Validate-Assignment.ps1 Named profiles: .\WIFIMGT-Validate-Assignment.ps1 -ProfileNames 'WIFIMGT - Corp WiFi - Windows - Pilot','WIFIMGT - Corp WiFi - iOS - Pilot' Sovereign cloud: .\WIFIMGT-Validate-Assignment.ps1 -Environment USGov (or USGovDoD) Required Graph scopes: DeviceManagementConfiguration.Read.All, Group.Read.All EXPECTED RESULTS Per profile: FOUND/MISSING, the assigned group name(s), and each group's member count. A profile with no assignment is flagged UNASSIGNED. .NOTES White-label — Intune Solutions. Read-only against Microsoft Graph. > [CmdletBinding()] param([ValidateSet('Global','USGov','USGovDoD')] [string] $Environment = 'Global', [string[]] $ProfileNames = @('WIFIMGT - Corp WiFi - Windows - Pilot', 'WIFIMGT - Corp WiFi - iOS - Pilot', 'WIFIMGT - Corp WiFi - Android - Pilot'))